Internet Language on Campus:

Linguistic Forms, Textual Patterns, and Educational Interpretation

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Abstract

Internet expressions are now a common part of student communication, but they differ widely in form and meaning. This project examines these expressions as a linguistic dataset, focusing on how features such as wording, length, script use, and cultural source relate to manually assigned categories. Descriptive analysis, text classification, and clustering are used to compare different expression types and identify recurring patterns. The findings suggest that some categories can be partly distinguished by visible textual features, while others depend more strongly on meaning, context, and cultural knowledge. These patterns offer a useful way to study how informal digital language enters campus communication and interacts with more formal language use.

1 Research Questions

Internet language changes rapidly because speakers continually reuse quotations, shorten phrases, create sound-based jokes, and attach new meanings to familiar forms. These expressions are therefore useful material for sociolinguistic study. They show how written Chinese interacts with alphabetic strings, cultural references, social events, and group-specific knowledge.

The campus setting gives this question an educational dimension. Students routinely move among social media, peer conversation, classroom discussion, and formal writing. Each context places different demands on vocabulary, tone, and convention. A useful research question is therefore how different internet expressions are formed and interpreted, and whether their linguistic categories can be recognized from the expressions and their usage contexts. Specificallyk, this project addresses three questions: (1) What observable linguistic differences appear across six manually defined types of Chinese internet expressions? (2) Can the expression and its example contexts recover part of the manual category information? (3) Do similar categories emerge when the same texts are grouped without access to the manual labels?

2 Data and Linguistic Representation

The analysis uses CHIME, a public dataset of Chinese internet expressions with meanings, origins, example sentences, linguistic types, and content-related labels [1, 2]. Related resources such as CHEER also show the value of studying internet buzzwords together with naturally occurring usage contexts [3]. Research on student social-media communication further motivates the educational setting in which such expressions circulate [4].

This study uses a balanced set of 18 verified expressions, with three entries in each of six categories: saying, rhetoric, quotation, phenomenon, abbreviation, and homophone. Each entry

contains three example sentences. Table 1 combines the validation output and the basic feature summary produced in the notebook.

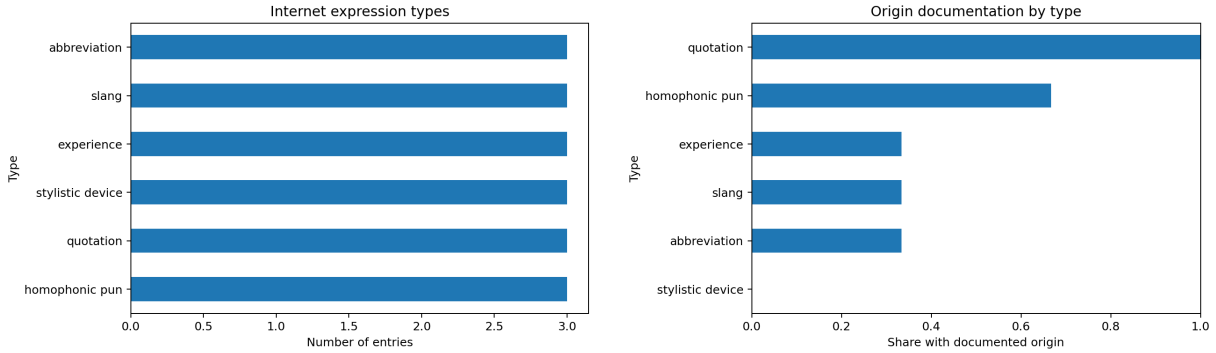
Table 1: Validated analysis sample

Measure	Value
Unique expressions	18
Linguistic types	6
Example sentences	54
Validation errors	0
Mean expression length (characters)	4.61
Share with documented origin	0.444

Each expression is converted into one analysis record. The representation keeps the expression, linguistic type, origin information, and example sentences, and derives several transparent features: character length, mean example length, whether an origin is documented, and the proportions of Latin letters and digits. These variables were chosen because they have direct linguistic interpretations. For example, Latin-letter share captures mixed-script forms, while origin documentation helps distinguish expressions tied to identifiable quotations or events from more general rhetorical constructions.

3 Descriptive Linguistic Analysis

The first analysis compares the six categories using counts and interpretable surface features. Figure 1 presents category frequency and origin documentation. The category plot confirms that the current sample is balanced, with three expressions in each type. The origin plot reveals a clearer linguistic contrast. All quotation entries have a documented origin, compared with two of three homophone entries and none of the rhetoric entries. This difference fits the nature of the categories: a quotation is normally connected to a source, whereas a rhetorical form can circulate without one identifiable point of origin.



(a) Number of expressions in each linguistic type.

(b) Share of expressions with a documented origin.

Figure 1: Category structure and provenance information in the analysis sample.

Table 2 summarizes the category-level statistics generated by the notebook. Phenomenon and homophone have the longest mean expressions, at 6.00 and 5.67 characters respectively, while rhetoric has the shortest mean at 3.33. Quotation has the longest example contexts on average

and complete origin documentation. Homophone is the only category with a substantial Latin-letter share, reflecting mixed forms in which alphabetic material participates in a sound-based expression. The content labels are sparse: one abbreviation and one homophone are marked as potentially offensive, and no sample item is marked for profanity.

Table 2: Linguistic features by manual type

Type	n	Share	Mean len.	Median len.	Mean ex. len.	Origin share	Latin share
Saying (俗语)	3	0.167	4.00	3.00	17.44	0.333	0.000
Rhetoric (修辞)	3	0.167	3.33	3.00	18.44	0.000	0.000
Quotation (引用)	3	0.167	4.33	3.00	23.89	1.000	0.000
Phenomenon (现象)	3	0.167	6.00	6.00	23.11	0.333	0.000
Abbreviation (缩写)	3	0.167	4.33	4.00	23.44	0.333	0.000
Homophone (谐音)	3	0.167	5.67	6.00	23.67	0.667	0.519

Figure 2 shows the distribution of expression length within each category. The plot complements the averages in Table 2: phenomenon and homophone extend toward longer forms, whereas rhetoric and saying remain concentrated at shorter lengths. The figure also illustrates why a single feature cannot define the categories. Length differs across types, but several categories overlap substantially.

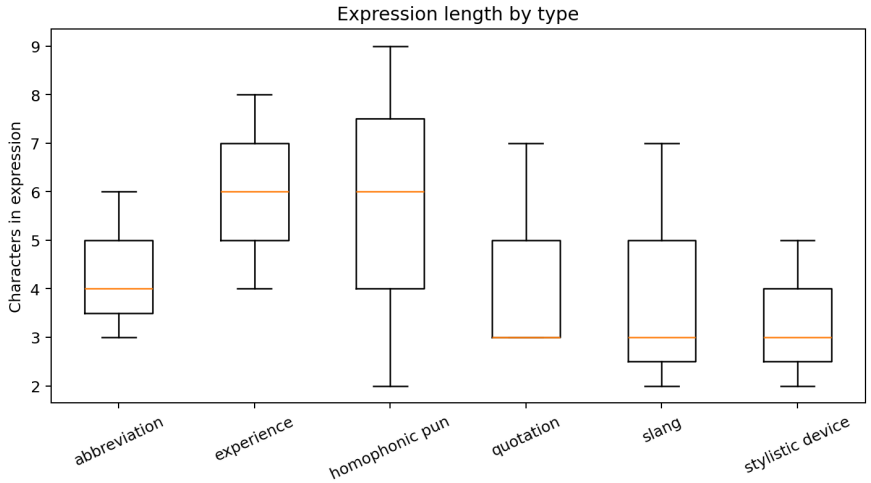


Figure 2: Expression length by linguistic type.

4 Simulation Analysis: Supervised Classification

The next stage asks whether manual types can be recovered from surface text. The model uses character n-gram TF–IDF features from the expression and its example sentences, followed by logistic regression. Character-level representation is useful for Chinese internet language because it treats Chinese characters, English abbreviations, and mixed forms within one feature space. Dictionary-style meanings are excluded from the model input so that explanatory definitions do not reveal the target label directly.

A stratified split places 12 expressions in training and six in testing, with one test expression from each category. Table 3 reports the resulting metrics. Accuracy reaches 0.333, above the 0.167 majority baseline, while macro-F1 is 0.194. The lower macro-F1 indicates that correct predictions are concentrated in a small number of categories.

Table 3: Character-level classification results

Metric	Value
Training entries	12
Test entries	6
Accuracy	0.333
Macro-F1	0.194
Majority baseline accuracy	0.167

The confusion matrix in Figure 3 shows where the model succeeds and fails. The saying and homophone examples are classified correctly. Rhetoric and quotation are predicted as saying, abbreviation is predicted as phenomenon, and phenomenon is predicted as homophone. The result suggests that surface form and example context carry some category-related information, especially for categories with distinctive writing patterns, but they do not reproduce the full manual taxonomy.

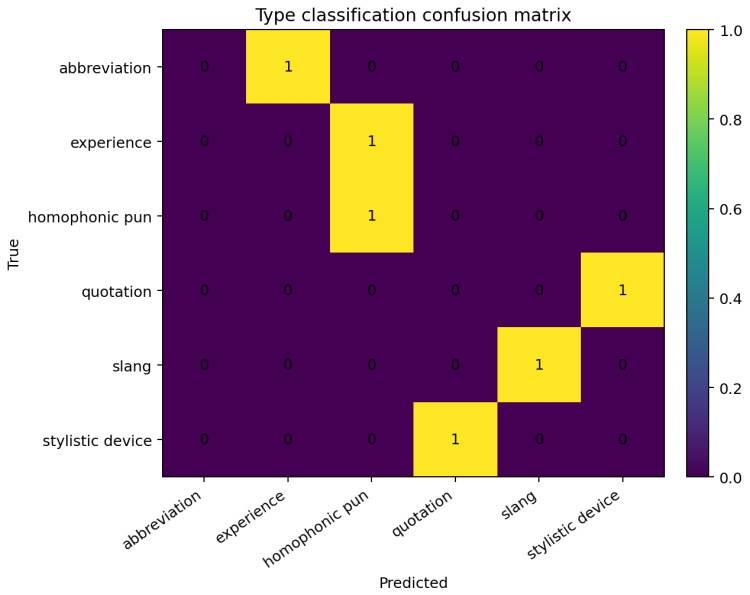


Figure 3: Confusion matrix for the character-level classification experiment.

5 Simulation Analysis: Unsupervised Clustering

The clustering stage removes the manual labels and asks whether the expressions form similar groups on their own. The same character-level surface text is reduced to a lower-dimensional representation and grouped into six clusters with k-means. Two statistics summarize the result: the silhouette coefficient measures how clearly the clusters are separated, while adjusted mutual information (AMI) compares the discovered clusters with the six manual categories.

Table 4: Unsupervised clustering results

Metric	Value
Number of clusters	6
Silhouette coefficient	0.0049
Adjusted mutual information	-0.2317

The silhouette value is close to zero, indicating weak separation among the six text clusters. The negative AMI shows that the discovered grouping does not positively align with the manual categories after chance adjustment. Figure 4 visualizes this mixing. The two largest clusters each contain expressions from five different manual types, while only one cluster contains a single type by itself.

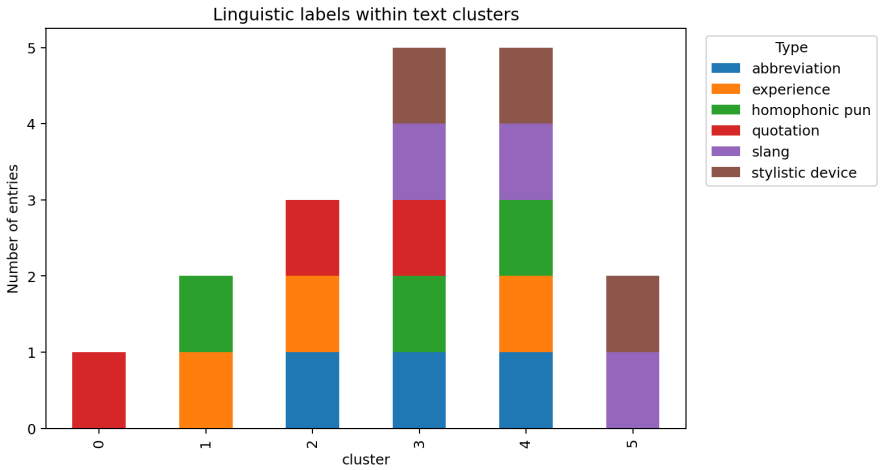


Figure 4: Manual linguistic labels within the six unsupervised text clusters.

Table 5 gives the numeric composition behind the figure. The mixed clusters help explain the low AMI. They also reveal an important linguistic point: the six labels describe different dimensions of language. Abbreviation is largely a form-based category, quotation is source-based, homophone is phonological, and phenomenon is semantic and cultural. Categories defined along different dimensions do not necessarily become separate geometric groups when only surface text is used.

Table 5: Cluster composition by manual linguistic type

Cluster	Saying	Rhetoric	Quotation	Phenomenon	Abbreviation	Homophone
0	0	0	1	0	0	0
1	0	0	0	1	0	1
2	0	0	1	1	1	0
3	1	1	1	0	1	1
4	1	1	0	1	1	1
5	1	1	0	0	0	0

6 Educational Interpretation

The descriptive and simulation analyses suggest that “internet language” should not be treated as one uniform style. Different expressions draw on different linguistic resources. Homophones

depend on sound relations and sometimes mixed writing systems; abbreviations compress written forms; quotations require knowledge of a source; and phenomenon labels package a social or cultural idea into a short expression. These differences matter when students interpret an expression or decide where it is appropriate to use it.

The gap between classification and clustering is also informative. A supervised model can recover some category information from surface text, but the categories do not emerge naturally when the same texts are grouped without labels. This means that understanding internet expressions requires more than recognizing spelling patterns. Meaning, cultural background, pragmatic intention, and shared community knowledge all contribute to interpretation.

For educational linguistics, this shifts attention toward register awareness. Frequent use of an abbreviation, quotation, or homophone in peer communication does not by itself indicate weak formal writing. A more useful question is whether a student can recognize the communicative setting, understand the expression's social and pragmatic meaning, and choose a more formal register when the situation requires it. The present analysis provides a concrete linguistic framework for discussing that ability.

7 Conclusion

This project turns familiar youth expressions into a structured linguistic research problem. The notebook moves from validated language records to interpretable features, descriptive comparison, supervised classification, and unsupervised clustering. Across the five figures and accompanying tables, the evidence shows both visible formal patterns and substantial overlap among categories.

The main finding is that internet expressions have measurable linguistic structure, but their categories are not determined by surface text alone. Their interpretation also draws on source knowledge, semantics, phonological relations, and pragmatic context. This layered structure explains why internet language is a productive topic for campus language research and for studying how students move between informal digital communication and formal educational settings.

References

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